

Operations Strategy Report: Transnet Port Terminals

TRANSNET



Improving Efficiency in the Operations of Transnet Port Terminals

Compiled by Syndicate 4: *FantasticFor*

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Executive Summary

Over recent years, Transnet Port Terminals, a Subsidiary of Transnet, has demonstrated itself to be in deviation of its fundamental business objectives. TPT, a state-owned entity, has rooted its business objectives in contributing positively to the economic growth of South Africa, reducing the associated cost with operating a business in the South African context as well as promoting itself to be customer-centric at its forefront. Yet, TPT currently finds itself in a position that operates in contrast to its articulated objectives.

Key Problems and Findings:

Our report has found that both TPT's operational bottlenecks and overall inefficiency has negatively impacted its ability to reach its business objectives. These bottlenecks and inefficiencies have been identified in particular facets of its operations in areas such as yard congestion, berth bottlenecks, system co-ordination failures, a hyperreliance on fragmented legacy ECC 6.0 system as well as financial mismanagement (Flynn et al., 2010). TPT's operational inefficiencies have contributed to an estimated R458 billion loss to South Africa's GDP, as of 2022 (Havenga et al., 2023). Additionally, these inefficiencies have strained the ease of operating a business reliant on the function of South African ports as well as fractured its customers' trust.

Recommendations:

We propose to implement a technologically-centric strategy that addresses the identified challenges whilst assisting TPT in realigning itself with its fundamental business objectives.

Our proposed strategy comprises of the following:

1. Incorporate the use Campbell Scientific AWS to detect adverse weather conditions prior to their disruption of port operations, in turn, mitigating any downtime.
2. Adopt Lean Synchronisation as a means to eliminate operational waste in areas such as waiting time, defects and overprocessing.
3. Incorporate Navis N4 in operations for greater yard and berth optimisation as well as utilising GCOS3 for real-time control.
4. Transition from legacy systems to SAP S/4HANA to ensure more uniformed data processing, predictive analytics and greater financial transparency.
5. Introduce performance-based contracts with Original Equipment Manufacturers (OEMs) to ensure asset reliability.

We are of the belief that the value and impact of our proposed strategy will be shown in the following:

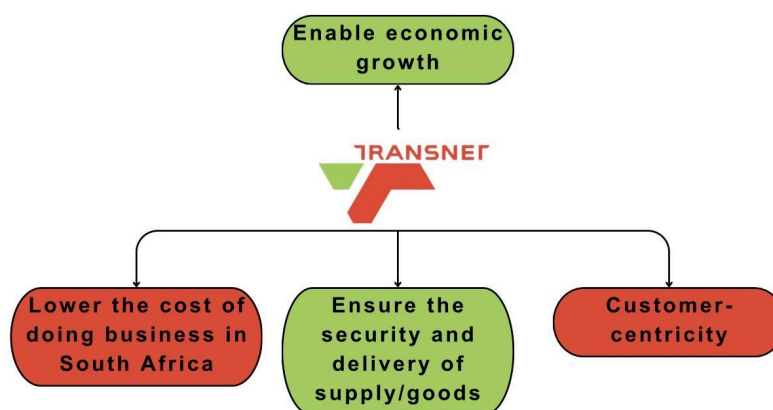
1. Allow TPT to regain better global competitiveness through greater trade efficiency and throughput.
2. Lead TPT to operate as a greener supply chain due to the reduction in maritime fuel emissions.
3. Improve TPT's ethical governance through greater financial transparency.
4. Operate with customer centricity through the provision of real-time data to its customers.

Operational efficiency at TPT is paramount to its future success. By integrating technology into its operations as a means to address its various challenges, TPT has great potential in reaching all of its business objectives.

Transnet History

The foundation of present day Transnet is rooted in 1872; with the South African government's acquisition of the Durban and Cape railway systems (Transnet, 2017). 1877 marked the beginning of South Africa's state railway system as it laid the foundation for the country's venture into enhancing its logistical infrastructure (Transnet, 2017). The importance of this pursuit was justified by the shifting economic landscape that saw an increased demand to move goods and people between emergent economic and mining hubs. In 1910, the government established the South African Railways and Harbours Administration (SAR&H) which later became the South African Transport Services (SATS) in the 1970s (Transnet, 2017). Throughout the process of its reconstruction, there has been a consistent emphasis on railways and harbour as a means to further develop the South African economy. SATS continued to evolve, reaching its present form as Transnet in 1990 (Transnet, 2023a).

Transnet in its current form is a South African state-owned enterprise positioned with the task and responsibility of operating South Africa's logistical infrastructure (Transnet, 2023a). Through its seven subsidiaries (Freight Rail, National Ports Authority, Port Terminals, Engineering, Pipelines, Property and Infrastructure Manager) Transnet manages extensive port, rail, and pipeline networks with its fundamental business objectives being rooted in: enabling South Africa's economic growth through the reduction of expenses associated with conducting business in South Africa, ensuring security of physical good and supply through the provision of appropriate infrastructure as well as being customer centric (Transnet, 2023a). Whilst Transnet's fundamental business objectives reign true to all seven of its



subsidiaries, it is to be noted that this report will focus solely on its subsidiary: Transnet Port Terminals.

Transnet Port Terminals Overview

Transnet Port Terminals is the division of Transnet responsible for the commercial handling of sea-route freight (Transnet, 2024). These terminals serve as the landing point for South African imports, and as the final handling point for exports (Transnet, 2023a). South Africa's geographical position largely informs TPT's operational strategy as Ports and terminals are to be positioned in a manner that enables regional, Eastern, European, American, and South-South trade access (Transnet, 2024). As of 2026, TPT operates 16 terminals across 7 South African ports in areas such as Richard's Bay, Durban, East London, Port Elizabeth, Ngqura, Cape Town and Saldanha (Transnet, 2024). For ease of operations, TPT categorises these terminals by geographic region (Eastern Cape, Western Cape and KwaZulu Natal) with each port having a unique combination of cargo and resources handled, as shown in Appendix A, (Transnet, 2024).

SIPOC Diagram Explaining TPT

SIPOC DIAGRAM

Transnet Port Terminals (TPT) — Core Port Operations

Process: Receipt, Handling, and Dispatch of Cargo Through South African Ports

S	I	P	O	C
SUPPLIERS	INPUTS	PROCESS	OUTPUTS	CUSTOMERS
• Shipping Lines (Maersk, MSC, CMA CGM) • South African Exporters (Agriculture, Mining, Automotive) • South African Importers • Original Equipment Manufacturers (OEMs) • CSIR (Weather Monitoring) • Road & Rail Transporters • SAP S/4HANA System • Transnet Freight Rail • Government & Regulators (PRSA, SAMSA) • Private PPP Partners	• Vessels & Cargo (Containers, Bulk, RORO, Break-bulk) • Port Infrastructure (Cranes, Conveyors, Berths) • Equipment (Straddle Carriers, RTG Cranes, Haulers) • Labour (Technicians, Operators, Planners) • Weather & Environmental Data (AWS) • Digital Systems (GCOS, EDI, SAP S/4HANA) • Maintenance Spares & OEM Contracts • Fuel & Energy Resources • Regulatory Permits & Documentation • Financial Capital & PPP Investment	• 1. Vessel Pre-arrival Planning & Scheduling • 2. Port Entry, Pilotage & Berthing • 3. Cargo Discharge (Imports) / Loading (Exports) • 4. Cargo Inspection, Weighing & Documentation • 5. Stack Planning & Terminal Storage Management • 6. Gate Processing (AGS/TAS — In & Out) • 7. Equipment Maintenance & Scheduling (SAP) • 8. Weather Monitoring & Risk Management (AWS) • 9. Reporting, Billing & Financial Processing • 10. Vessel Departure & Performance Review	• Cleared Import Cargo to Domestic Market • Loaded Export Cargo to International Markets • Vessel Turn-around Records & Performance Data • Port Throughput Volumes (TEUs, tons, units) • Real-time Operational & Financial Reports (SAP) • Equipment Maintenance Records • Weather Risk Alerts & Contingency Plans • Customs & Regulatory Compliance Documentation • Revenue (Handling Charges, Tariffs, Fees) • Supply Chain Performance Metrics	• South African Importers & Exporters • Shipping Lines & Vessel Operators • Landlocked SADC Countries (Zimbabwe, Zambia, Botswana, etc.) • South African Agricultural & Mining Sectors • Automotive Industry (Ford, Toyota, BMW) • Transnet Group (Freight Rail, Pipelines) • Port Regulator of South Africa (PRSA) • South African Government & National Treasury • Private PPP Concession Partners • South African Economy (GDP & Trade Competitiveness)

Source: Transnet Port Terminals Operations Strategy Report (2026). Compiled with reference to Transnet Integrated Annual Reports, Port Regulator of South Africa, and proposed TPT operational strategy.

Current Operations of Transnet Port Terminals

Key technologies currently used in Transnet's Port Terminals include, General Cargo Operational System (GCOS) as well as Electronic Data Interchange (EDI)(Transnet, 2023b). General Cargo Operational System (GCOS) is the system utilised for the operational and administrative processing at Transnet Port Terminals (Transnet, 2023b). This web based

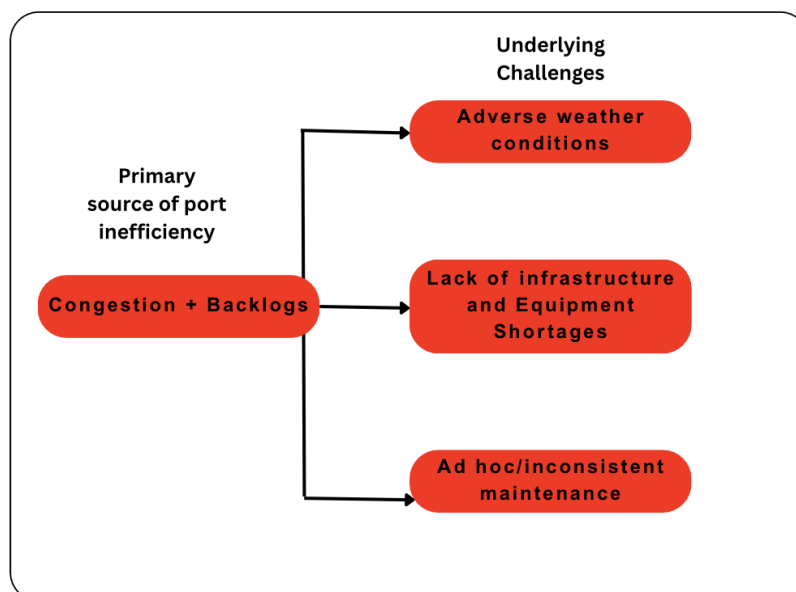
system features: Real-time data managed on one database, reports and query management at any point in the operation, Electronic Data Interface, Graphical representations of moving cargo (Transnet, 2019). The processes supported by this system include: Vessel planning, Import and Export Administration, Booking lists that are done both done manually or electronically, Stack planning for the optimisation of space at terminals, Weighbridge whereby cargo is managed by weight, Cargo tracking through handheld cargo scanners as well as Financial reports (Transnet, 2023b).

Secondly, the Electronic Data Interchange (EDI) is utilised for the reporting of cargo movements. EDI is the inter-organisational, computer-to-computer exchange of business information in a standard, machine processible format (Transnet, 2023b). Processes supported by the EDI include: Equipment Release Orders, Equipment Acceptance Orders, Gate-Out Confirmations, Gate-In Confirmations, Vessel Load and Discharge Confirmations and Storage instructions (Transnet, 2023b).

At present, Transnet Port Terminals' occupy a vital position in South Africa's economic growth. However, due to numerous recurring operational challenges there has been a hindrance in its ability to achieve its articulated business objectives.

Transnet Port Terminals (TPT) Challenges:

Challenges that currently plague Transnet's Port Terminals (TPT) can be attributed primarily to the ports' inefficiencies. The ports' have been noted for having extensive congestion and backlogs in its operations. This is due to factors such as adverse weather conditions, a lack of infrastructure, shortage of appropriate equipment as well as inconsistent maintenance of its respective infrastructure.



All Cargo Handling	Terminal area (sqm)	Theoretical Capacity	Installed Capacity	Volumes Handled	Excess Capacity
Automotive	681 022	2 100 000	833 200	700 514	132 686
Container	2 031 605	7 150 000	5 471 955	4 619 623	852 332
Dry bulk	1 755 002	198 100 000	195 084 000	150 409 450	44 674 550
Break bulk	2 240 103	15 420 000	14 997 000	15 899 906	-902 906
Liquid bulk	2 821 537	414 600 000	39 020 000	23 495 020	15 524 980

*These insights from Port Regulator SA (2024) reflect the true extent of the port inefficiencies within Transnet as it highlights consistent latent capacity in all cargoes handled except break-bulk which faced shortages and lack of storage.

1. Digital Challenges in Transnet's Port System

Transnet's port system has historically relied on ECC6.0, a fragmented legacy system, to manage finance, procurement, asset maintenance, human resources, and supply chain processes (Flynn et al., 2010). ECC 6.0 is a disconnected system which limits real time visibility across operations (Flynn et al., 2010). This contributes extensively to inefficiencies such as delayed procurement, inefficient maintenance planning, inaccurate inventory management, and weak financial control. Operational disruptions at major ports have often been linked to both physical infrastructure challenges as well as planning and coordination failures (Flynn et al., 2010). The lack of integrated information systems has weakened capacity planning, inventory control, demand forecasting and asset utilisation. Thus, digital transformation proves pivotal in the stabilisation of overall operational system performance (Flynn et al., 2010).

2. Mismanagement:

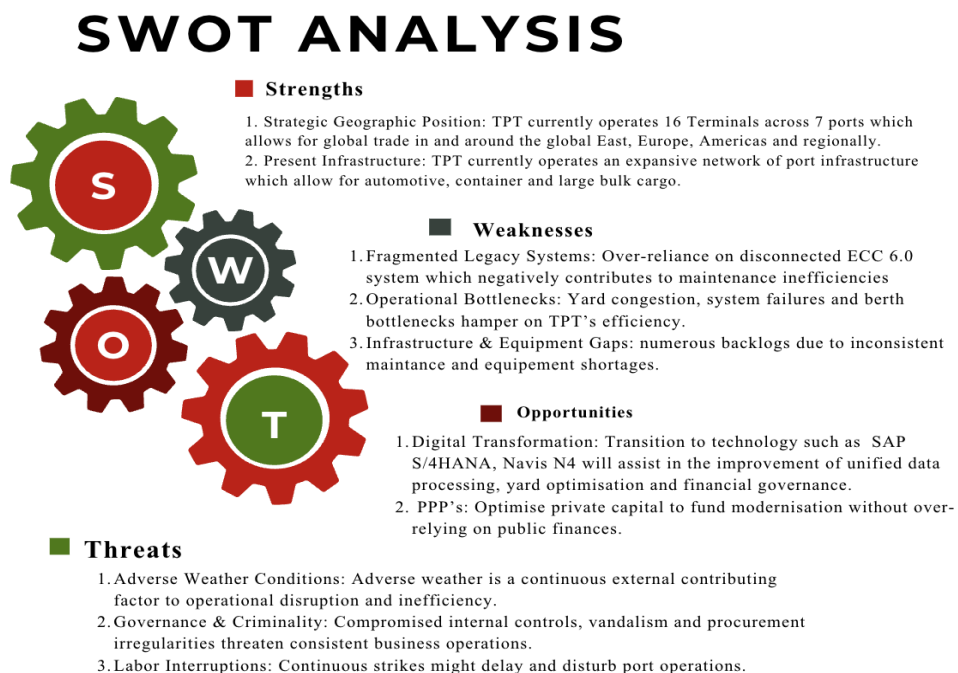
Governance and Management presents itself as a challenging aspect as fragile internal controls, procurement irregularities and poor financials all reporting has directly contributed to undermined performance and compromised accountability (Dean, 2024). The legacy of corruption and the impact of state capture has left the state-owned entity vulnerable to hampered institutional capacity (Dean, 2024). Moreover, Transnet has navigated challenging negotiations in the past and thus always has a posing threat that strikes will disrupt and delay its operations (Dean, 2024). In addition, theft, vandalism and security issues are chronic problems that disrupt services (Dean, 2024). Criminal activity has significantly impacted procurement and operations. Thus, highlighting the need for adequate measures, control systems and management processes.

TPT's Division Financial Performance

Despite group-level losses driven primarily by Freight Rail and Engineering, TPT has remained operationally profitable. However, the absolute figures mask the structural fragility (See Appendix B) as these figures prove to be misleading in isolation. The South African Port Regulator has repeatedly noted that Transnet cross-subsides its divisions with losses (e.g Freight Railway) with profits extracted from TPT's operations (Havenga et al., 2023; Chasomeris & Fakir, 2020). This results in a loss of Capital for TPT operational activities and investments. Transnet Port Terminals' operational inefficiencies have negatively impacted South Africa's economy (as shown in Appendix C). According to (Havenga et al, 2023), the total infrastructure costs to the economy increased from R84.5 billion (1.5% of GDP) in 2019 to R458.4 billion (6.7% of GDP) by 2022. At the business and sectoral level, the financial implications have both been acute and structural. The agricultural sector bore some of the most visible losses: the fruit export industry lost R2 billion in the 2022/23 financial year (Transnet, n.d)). Additionally, TPT's own estimates and industry reporting reveal the direct financial haemorrhage from operational failures (See Appendix D). Thus, these losses represent not merely a logistics inconvenience but a structural tax on TPT, South African business competitiveness, and overall supply chain. Pertaining to the supply chain, Maersk, CMA CGM, and MSC imposed Durban port congestion surcharges, increasing landed costs for importers and reducing export competitiveness (Mthembu & Chasomeris, 2025). Additionally, neighbouring landlocked countries (Zimbabwe, Zambia, DRC, Eswatini, Lesotho, Botswana) that transit goods through South Africa's ports face knock-on delays and increased logistics costs. As the data shows there are massive financial costs to TPT's operational inefficiencies.

SWOT Analysis: Transnet Port Terminals

Based on the identified challenges above, the following SWOT analysis image summarises the overarching strengths, weaknesses, opportunities and threats TPT faces.



Current Operational Strategy: Transnet Recovery Plan (2023)

Due to the numerous challenges negatively impacting Transnet's operations, as mentioned above, a recovery plan using a two-pronged approach was first introduced in 2023. The first facet addresses issues of optimisation and enhancement by focusing on recovering volumes lost due to previous inefficiencies (Transnet, 2023b). The second aspect regards the need for organisational transformation within Transnet (See Appendix E).

In relation to TPT, the optimisation of operations would be through: the deployment of additional rolling stock at strategic points, Increased container moves per ship working hour across all terminals as well as the establishment of Operations Excellence Centres (OEC) to foster more collaboration and increase volume delivery. Moreover, The recovery plan suggests maintenance would be improved through the availability of spares to support turnaround time as well as the Establishment of partnerships with Original Equipment Manufacturers (OEMs) for technical support (Transnet, 2023b). Lastly, procurement would be optimised through the automation of process elements, the adoption of digital technologies and external support for clearing backlogs as per the Recovery Plan.

Proposed Operations Strategy

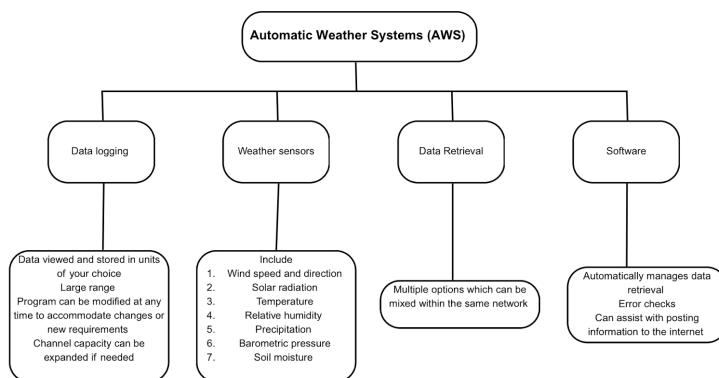
Whilst Transnet has begun the implementation of the Recovery Plan we propose an alternative operations strategy as a means to better assist Transnet in achieving its fundamental business objectives. This proposed operations strategy is rooted in the integration of technology as a means to address the fundamental contribution to the inefficiency of TPT: that being the congestion and overarching operational waste associated with TPT. Our proposed strategy aims to address TPT's blockages and operational waste through the provision of effective weather detection systems, lean synchronisation to address waste and maintenance as well as technological advancements in various facets of the subsidiaries operations.

1. Weather Detection Systems

Adverse weather conditions are continually cited as a key operational challenge as they impinge on the effectiveness of equipment and largely contribute to congestion at all stages through the ports. According to UN Trade and Development (2025), weather issues at ports should be dealt with both proactively and reactively, with primary emphasis on the proactive approach. Transnet, however, diverges from this advisory in that its primary focus is on crisis management after operations have been disrupted. Whilst TPT does have an early weather detection system integrated into their operations, its scope is limited, monitoring only wavelength and wind speeds/direction. Moreover, much of the monitoring is done by CSIR on behalf of Transnet and relevant weather information needs to first be collected, then transmitted, then analysed, and finally relayed (CSIR, 2018). Whilst these insights are helpful for pre-empting potential weather events, the various stages through which the information has to go creates delays that can hinder proactive risk management planning. As the effects of climate change weigh heavily, weather detection systems need to be comprehensive to

foresee events that our shore may have never experienced. Prior to the KwaZulu -Natal floods in 2021, a relatively simple system would've sufficed but as instances of flooding and high wind speed, and generally unpredictable climate start to increase its vital adopts a future-focused system that accounts for all possible weather events.

With this contextualisation, our operations strategy proposes the use of Campbell Scientific Automatic Weather Stations (AWS). Campbell's AWS is an internationally reputable system that specialises in meteorologic instruments. Its systems integrate virtually all aspects of forecasting and monitoring, and is entirely customisable to the climate and context of the client's needs (Campbell's Scientific, n.d).



Sourced from: Campbell Scientific (2026)

Potential benefits to TPT's operational efficiency include: ability to integrate into the SAP rollout, alignment with international standards, Informs risk management procedures and allows for appropriate planning to avoid congestion at ports due to adverse weather conditions, data can easily be relayed to customers, thus allowing expectations to be adjusted accordingly. This would positively impact customer satisfaction and ultimately maintains the core value of customer centricity. Additionally, Data analysis of trends over time can allow more informed equipment purchases.

2. Maintenance and Waste

TPT is understood as a mass service process type as it has many customer transactions, involving limited contact time and little customisation (Slack & Brandon-Jones, 2019). Such service is heavily reliant on equipment and is 'product-oriented', with most value added in the back office with little judgement applied by front-office staff (Slack & Brandon-Jones, 2019). Staff are likely to have a closely defined division of labour and follow set procedures (Slack & Brandon-Jones, 2019). This correlates with Transnet Port Terminals' daily operations as it involves the use of equipment to process clients' orders, prepare goods to be sent out as well as reduce time spent instructing employees (Slack & Brandon-Jones, 2019). For Example: Example is receiving cranes from the incoming ships, placing the cranes in a section where they will be transported to their next destination. There's little time to be

customising the goods exactly how the consumer wants it as the business is dealing with mass inflow of goods that need to move from one place to another.

Lean Synchronisation

Lean synchronisation aims to meet demand instantaneously, with perfect quality and no waste (Nigel Slack, 2010). The incorporation of lean synchronisation is fundamental for efficient proceedings at the Transnet Port Terminals as lean synchronisation is founded on smoothing flow through processes by doing all simple things well, gradually doing them better and ultimately extracting waste at every step of the way (Nigel Slack, 2010). Lean synchronisation aims for continuous improvement which will be beneficial for TPT as it ensures systems are always updating and will reduce the amount of backlogs experienced at ports, waiting time, congestion at port terminals, equipment breakdowns, inefficiencies in rail and logistics operations, leaving cargo ships waiting offshore for days and slowing exports of key commodities which will improve efficiency at TPT and boost its business image after experiencing a heavy knock to its image due to the inconsistencies experienced at the ports over the past years.

Elimination of waste and the seven types of waste

The most significant part of the lean philosophy is its focus on the elimination of all forms of waste (Nigel Slack, 2010). The seven types of waste which have been found to apply in many different types of operations, both service and production and which form the core of lean philosophy have been identified by Toyota (Nigel Slack, 2010): Over-production, Waiting time, Transport, Process, Inventory, Motion, Defectives. Overprocessing waste occurs when more work is done than necessary or when tasks must be repeated due to errors or equipment failure. In TPT, this type of waste is common when poorly maintained cranes and terminal tractors malfunction and require containers to be handled more than once.

Inventory waste is, “the excess materials, spare parts, or work-in-progress that are not immediately required” (Singh & Singh, 2021). In port terminals, poor maintenance practices often lead to equipment breakdowns that create container backlogs in storage yards. When cranes or straddle carriers are not operational, containers accumulate and occupy valuable terminal space, which limits yard capacity and slows down cargo movement. In addition, frequent machine failures encourage the overstocking of spare parts as a precautionary measure, tying up financial capital in unused inventory. This form of waste increases storage costs and reduces operational flexibility. Research on lean supply chains shows that excessive inventory is a direct result of unreliable equipment and unplanned downtime, and it weakens overall supply chain performance (Singh & Singh, 2021).

Defective waste is when outputs do not meet quality standards and must be corrected or replaced (Singh & Singh, 2021). These types of waste have a way of leading to other wastes. In port terminals, defects are often caused by poorly maintained handling equipment such as cranes, forklifts, and spreaders. Faulty machinery may drop containers, damage cargo, or load containers incorrectly onto vessels or trucks. These defects require rework, inspections, and

sometimes compensation claims from customers, all of which increase operational costs. For example, damaged containers must be removed from the process and reloaded, delaying shipments and disrupting schedules. Harrington and Wang (2020) explain that defects generate both direct costs, such as repairs and replacement, and indirect costs, such as loss of customer confidence and reduced productivity.

Lastly, Waiting waste is the idle time when workers, equipment, or cargo are unable to proceed to the next stage of operation (Singh & Singh, 2021). Poor maintenance creates waiting waste when machines break down unexpectedly and operations are forced to stop. For instance, ships may wait at berths because cranes are under repair, while truck drivers and terminal workers remain idle until equipment becomes available again. Staff continue to be paid even though productive work cannot take place, which results in financial loss. This waste also increases congestion and backlog in terminal operations. According to Singh and Singh (2021), waiting time is one of the most visible consequences of equipment unreliability and weak maintenance systems in logistics operations.

Maintenance can be understood as, “The process of preserving a condition or situation”(Oxford Languages, n.d). The primary purpose of maintenance is to keep equipment working properly and to extend its lifespan. Effective maintenance assists in the prevention of challenges such as congestion, delays, and backlogs in operations (Mobley, 2011). Research shows that maintenance helps all parts of a business and its assets to work as well as possible.

However, maintenance is multifaceted and requires meticulous planning as there are varied forms of maintenance. Mobley explains different maintenance methods, such as preventive maintenance, predictive maintenance, and repair (Mobley, 2011). Preventive maintenance includes regular tasks like lubrication, adjustments, and rebuilding important machines. Predictive maintenance means checking and monitoring machines to find problems early and prevent serious breakdowns before they happen. Using a system that monitors maintenance activities can help to ensure high equipment availability, better performance, longer equipment life, fewer spare parts needed, and faster response to failures.

Transnet has introduced SAP S/4HANA to improve operational efficiency, improve how data is processed, and support the use of machine learning (Moyo, 2025). In addition, our operations strategy suggests that transnet could adopt performance based maintenance contracts with specialised engineering service providers as part of its operations strategy. This approach would help address the challenge that inspections are often carried out by “inexperienced Engineers and Technicians” and are limited to basic visual assessments (Mneney, 2015). Under performance-based contracts, external maintenance teams would be held responsible for reducing equipment downtime and preventing frequent breakdowns rather than only repairing damaged assets. This would shift maintenance away from reactive practices, which have been identified as a major weakness in Transnet’s operations (Mneney, 2015,). Such contracts could also include skills transfer and training requirements, allowing Transnet staff to learn better maintenance practices from experienced professionals. By linking payment to asset reliability and performance outcomes, Transnet would improve the

quality of maintenance work, strengthen accountability, and ensure that infrastructure is preserved more effectively over time (Mobley, 2011).

3. Technology Integration: SAP S/4HANA

Digital transformation is a vital component of the broader operational reform within Transnet's port system. The digital modernisation and transformation of Transnet's software system is vital for many of the proposed improvements within Transnets. A key strategy within this transformation agenda would be the implementation of SAP S/4HANA across Transnet's operations. SAP S/4HANA is an advanced enterprise resource planning system.

An ERP (Enterprise Resource Planning) is a software system that informs data collected from many sub-sources into a single system (Oracle, 2020). SAP S/4HANA is SAP's latest ERP software suite, built on the company's HANA in-memory database platform, enabling real-time data processing, advanced analytics, and integration of intelligent technologies like artificial intelligence (AI) and machine learning (Moyo, 2025). SAP aims to modernise enterprise resource planning processes, integrate operational data, and enhance decision making across the organisation. It is to be noted that Transnet has initiated the Integration of this technology throughout, including port operations managed by Transnet National Ports Authority and Transnet Port Terminals (Transnet, 2019).

a. Predictive Maintenance

SAP S/4HANA's most significant operational benefits relate to asset management. Port operations depend heavily on high value equipment such as conveyor systems, ship to shore cranes and straddle carriers. Equipment breakdowns have historically contributed to vessel delays and congestion thus resulting in backlogs. SAP enables preventive and predictive maintenance scheduling through integrated maintenance modules (Tsang, 2002). It enables maintenance teams to track asset performance, record faults, and schedule services in real time. This improves equipment reliability and reduces unplanned downtime (Tsang, 2002). In turn, this would result in an increase in throughput and it resultantly stabilises service delivery. Adequate asset management thus supports a better lifecycle of costing assets and leads to better informed data driven capital investment decisions. (Tsang, 2002)

b. Procurement and Supply Chain Integration

Internal procurement systems which operated under fragmented legacy systems resulted in poor coordination and lack of integration. SAP functionally integrates the procurement processes resulting in an improved and coordinated procurement system. Automation is central in the functionality of the workflow thus reducing timely manual systems. Moreover, it improves tracking and accountability. Port operations rely heavily on efficient sourcing thus automation improves functionality. Additionally, it enhances supplier management and contract compliance, contributing to cost control and operational continuity.

c. Data Driven Decision Making

SAP S/4HANA's in memory capabilities enable real time analytics and dashboard reporting (Grabski et al., 2011). Thus, operational managers access up to date information on asset

performance, procurement status, expenditure, and operational metrics. This resultantly reduces reliance on outdated reports and manual data consolidation. Instead of reactive problem solving, management can identify trends, demands, and respond proactively to emerging challenges. Moreover, Integrated data facilitates alignment between port operations and end to end supply chain coordination.

4. Technology Integration: NAVIS N4

Within Transnet Port Terminals, Navis N4 supports berth planning, crane scheduling, yard allocation and gate operations. One of the primary operational challenges at South African ports has been vessel backlogs and extended waiting times. From an operations management perspective, long waiting times represent a form of process waste that reduces system efficiency and increases costs. Navis N4 enables planners to allocate berths more effectively by providing real-time information on vessel arrival, container load plans and crane availability. This improves scheduling accuracy and reduces idle time for both vessels and equipment (Kaizen). In addition, Navis N4 enhances yard management. Container yard congestion is a common bottleneck in high-volume terminals. Poor yard planning can result in unnecessary container reshuffling, increased equipment movement and longer truck turnaround times. Through automated yard planning algorithms and real-time container tracking (Just-In-Time), Navis N4 improves stack optimisation and reduces unnecessary handling.

a. Performance monitoring.

Terminal managers can access dashboards that display crane productivity rates, berth occupancy levels and equipment utilisation metrics. This supports data-driven decision-making and continuous improvement. Rather than reacting to congestion after it occurs, managers can identify emerging bottlenecks and intervene earlier. In this way, Navis N4 contributes to a more proactive operations strategy within the port terminal environment.

Complementing Navis N4 is GCOS3 (Global Container Operating System version 3), which focuses on container tracking, documentation integration and system-wide coordination. While Navis N4 manages operational execution on the ground, GCOS3 enhances information accuracy and interdepartmental communication. Accurate documentation and real-time container status updates are essential for reducing administrative delays, customs hold-ups and cargo misallocations. Christopher (2016) highlights that visibility across the supply chain is a key driver of efficiency and customer service performance. In the port terminal context, GCOS3 improves transparency from vessel discharge to container release.

Technology and Sustainability:

As highlighted in our strategy the use of SAP S/4HANA and AWS technology acts as a central figure in the overall efficiency of TPT. The introduction of technical transformation in TPT would further allow for a positive impact in its sustainability efforts as SAP S/4HANA allows for an integrated materials management function (Annanki, 2025a). This has the potential to provide limitless data in the resource consumption patterns spread throughout TPT's production processes, thus, accurately and effectively recognising segments in which

TPT can reduce their overall waste (Nidumolu, Prahalad, 2024). Energy management systems help monitor overall consumption, timeously, which helps inform schedules and strategies set in place to reduce both carbon footprints and expense (Annanki, 2025a).

Technology and Ethics

Financial Transparency and Governance

Transnet operates under strict regulatory and governance requirements as it is an SEO. Accountability is vital as financial mismanagement, irregular expenditure and poor operational governance continue to undermine the organisation's credibility. Technological integrations within the TPT's operations would ensure a centralised, digitised and transparent internal control system. This would reduce the ability of financial mismanagement, misuse of company funds or malpractice within the organisation as there would be greater and more robust auditable systems in place, tracking all financial actions. Additionally, the integration of technology within the administrative and financial management of TPT could ensure ethical collaboration with PPP through traceable and auditable transactions and selection of third party entities as opposed to easily misplaced, or untraceable manual of physical records of such transactions.

Upskilling

With the integration of digital transformation, TPT would need to place active organisational efforts in the upskillment of its current employees to both maintain an effective and efficient workforce as well as abide by ethical principles. Research has shown that the integration of SAP/4HANA and other technological investments within an organisation results in significant multiplier effects with the upskillment of organisational workers (Annanki, 2025a). The rise and use of technology within an organisation fluidly correlates with the uptake in upskilling workers in areas such as user experience design, ongoing support services and implementation management (Wipro, n.d). Additionally TPT, and more broadly Transnet, taking intentional efforts to upskill workers with technical skillsets such as cloud analytics and data analysis would result in a more robust, agile and effective workforce whilst ensuring a greater use of the technical facilities incorporated in the entity. Such programmes would ensure better planning and scheduling, faster communication amongst staff and the organisation entirely, good management of inventory, and potential unity across all the departments.

Respecting Human Capital

TPT would counter the fear associated with Lean could counter this through intentional and conscious employment. The maintenance of human labour within the technological integrations imposed in our strategy acts pivotal in TPT ensuring an ethical practice within TPT operations. Ethical and Intentional maintenance of human labour within TPT would be through the employment of low-wage physical labour in communities with limited financial opportunity outside of TPT.

Corporate Social Responsibility

In its current form TPT, and Transnet more broadly, have existing corporate social responsibility efforts filtered through the Transnet Foundation. The foundation's CSR is positioned as human-centric with an emphasised focus in areas of broad community health, education and wellbeing (Transnet, n.d). At present Transnet's CSR is positioned as separate and supplementary to its business operations. However, the introduction of SAP S/4HANA within its CSR efforts could result in a broader integration and width-net impact of its CSR efforts. The use of SAP S/4HANA would allow for greater transparency in its CSR impact measurement through analysing its social performance indicators parallel to financial metrics by way of its comprehensive data capabilities (Annanki, 2025a). This would allow for greater understanding of the efficiency, efficacy, implementational strengths and shortfalls of any and all CSR initiatives (Annanki, 2025a). Ultimately, contributing to an optimised CSR strategy. This could result in a greater focus and investment in CSR initiatives that prove to be fruitful and impactful to society more broadly in the short, medium and long term. Contrastingly, to randomised, manually selected initiatives that might have greater potential to be futile in the medium to long-term.

Total Quality Management

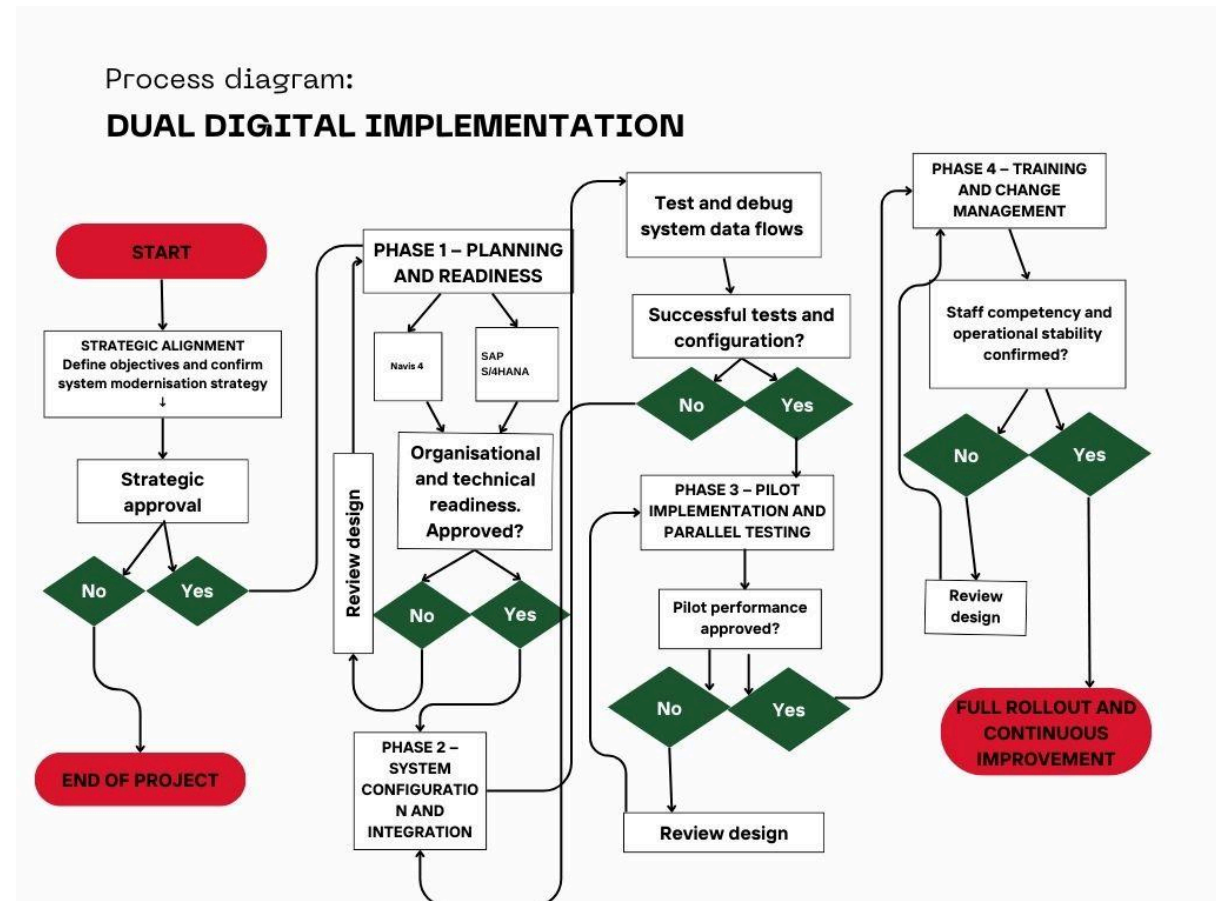
The Total Quality Management of TPT would be positively impacted by the integration of technology. This, in turn, results in a positive impact in the sustainability and ethical efforts of TPT. Total Quality Management is understood as a customer centric philosophical framework centered on the notion of collective and cohesive collaboration amongst all employees within an entity as a means to improve business processes, products and services (Bamford, Dale & Wiele, 2016). In the context of TPT, the introduction and implementation of SAP S/4HANA and Navis N4 (as suggested in our strategy), could result in the reduction of TPT's carbon footprint, an introduction of a green supply chain as well as a reduction in unethical financial mismanagement. Additionally, a focus on intentional upskilling, an effort to respect human capital and corporate social responsibility further act as suggestions to maintain ethical practices within TPT whilst maintaining alignment with its overarching business objectives and leading to long-term sustainability.

Implementation Plan

Technological

Our strategic objective aims to modernise both port and enterprise systems, improve Efficiency, reduce waste, and improve vessel turnaround times, ensuring accurate billing financial control, and providing management with real time visibility into operations. In sum, the port floor and the finance office should be operating on the same reliable data (Arena et al., 2020). One of the biggest challenges is complexity. That being, integrating an enterprise system with an operational terminal system, in a high-volume, high-pressure port environment. If this is rushed it can disrupt operations. Thus, the strategy focuses on control and structure. First, we use a phased rollout approach. We avoid a simultaneous launch as that creates confusion and disorder. Instead, the strategy is implemented in stages. Secondly, a

dual track implementation structure is followed. While these systems are implemented in parallel, a central priority is ensuring seamless integration between them. Additionally, the operational systems can thus be agile and enable informed decisions that are based on big data (Goswami, 2020). A successful digital transformation implementation plan is realising a well-defined approach (Ayyalasomayajula, 2022). The implementation happens in four key phases.



Phase 1: Planning and Readiness Highlighting Objective Settings. This includes mapping current processes, cleaning data, identifying risks, and confirming infrastructure readiness.

Phase 2: data readiness highlighting system configuration and integration, where both systems are tailored to operational needs and connected properly.

Phase 3: Pilot implementation and parallel testing. The new system is run alongside the old one to reduce risk and confirm accuracy.

Phase 4: Training and Change Management.

Staff must understand the new processes, and management must actively support adoption to ensure long-term success. This strategy prioritises stability, incremental deployment and disciplined execution. Thus assisting Transnet in modernising its systems without disrupting operations and improving overall performance.

Financial:

a. Protect Port Revenues

Establish a legally protected TPT Capital Reserve Fund where a minimum of 35% of EBITDA (approximately R2.2bn based on 2024 EBITDA of R6.33bn) is ring-fenced annually for port maintenance and capital expenditure. This prevents continued cross-subsidisation of Freight Rail and Engineering and directly addresses the maintenance underspend gap identified by the Journal of Transport and Supply Chain Management (2023).

b. Emergency Equipment OEM Contracts

Accelerate and expand the 10-year OEM contracts initiated in April 2023 across all major terminal equipment categories (ship-to-shore cranes, rubber-tired gantry cranes, mobile harbour cranes, trailers, haulers). These long-term contracts, structured as operating leases or hire-purchase facilities, transfer equipment lifecycle risk to OEM suppliers while ensuring guaranteed maintenance and spares availability. Estimated annual cost: R1.5-2.5bn, offset by reduced breakdown-related revenue losses.

c. Terminal Concession Programme

Introduce competitive, long-term terminal operating concessions (20–35 year terms) at selected terminals, beginning with Durban Container Terminal Pier 2, where Transnet has already invited private sector proposals for the North Quay reconstruction. The concession structure should follow the COSCO-Piraeus model: the concessionaire funds all superstructure and equipment capital expenditure in exchange for long-term operating rights and revenue share. Financial modelling based on Piraeus' precedent suggests concession proceeds of R8-15bn upfront and annual concession fees of R1.5-3bn, depending on terminal scale and traffic projections. The private operator's capital injection (estimated R5-12bn per major terminal for equipment and technology) comes entirely off the public balance sheet.

d. Public-Private Partnerships as a Financial Strategy for Transnet Port Terminals

Public–private partnerships (PPPs) have emerged as an essential financial strategy for infrastructure development in capital-intensive sectors such as port and freight logistics. Freight transport infrastructure plays a critical role in enabling trade, improving supply chain efficiency, and supporting national economic growth (Ittmann, 2017). However, infrastructure expansion requires substantial capital investment, and state-owned enterprises such as Transnet face financial constraints that limit their ability to independently finance modernisation and expansion (Havenga et al., 2023).

Financial Structure and Capital Investment Benefits

One of the primary financial benefits of PPPs is the ability to fund infrastructure expansion through private sector investment. For example, Transnet National Ports Authority entered into a concession agreement with the Grindrod Eyamakhosi Joint Venture to develop a container handling facility at the Port of Richards Bay, involving private sector investment of approximately R285 million (Engineering News, 2025). This investment increases the port's container handling capacity from 50,000 to 200,000 twenty-foot equivalent units annually, significantly improving infrastructure utilization and revenue potential (Engineering News,

2025). Financially, private capital injection reduces Transnet's capital expenditure requirements and preserves internal financial resources. This improves Transnet's cash flow position and allows it to allocate financial resources to other operational and strategic priorities. PPPs therefore improve Transnet's capital structure by reducing reliance on debt financing and public sector funding.

Revenue Growth and Asset Utilisation

PPPs contribute directly to revenue growth by increasing infrastructure capacity, improving operational efficiency, and increasing throughput volumes. Higher throughput allows Transnet to generate increased revenue through terminal handling charges, port tariffs, and concession fees. For example, the Richards Bay container facility concession improves operational efficiency through modern equipment and infrastructure, enabling faster cargo handling and improved vessel turnaround times (Engineering News, 2025). Increased efficiency enhances the financial return generated from infrastructure assets and improves overall financial performance. Similarly, Transnet has issued a tender inviting private partners to participate in the development and operation of the Richards Bay commodities terminal. This PPP structure involves private partners contributing financing and operational expertise through a special-purpose vehicle to increase export capacity and improve operational performance (News24, 2026). Increased export capacity improves revenue generation potential by enabling higher cargo volumes to be processed through the terminal. Improved infrastructure performance enhances asset utilization, increasing the return on infrastructure investment and improving long-term financial sustainability.

Financial Risk Reduction and Investment Efficiency

Infrastructure development involves significant financial risks, including construction risk, operational risk, and revenue uncertainty. PPP arrangements reduce Transnet's financial exposure by sharing these risks with private sector partners (Ittmann, 2017). Risk sharing improves financial stability by reducing the financial burden associated with infrastructure investment while allowing Transnet to benefit from improved infrastructure performance.

The partnership to improve operational efficiency at Durban Container Terminal Pier 2 demonstrates how PPPs enhance infrastructure performance while reducing financial and operational risk. This concession agreement enables private sector participation in terminal operations, improving productivity and efficiency at one of South Africa's most important container terminals (SAnews, 2025). Improved efficiency increases throughput capacity and revenue generation while reducing operational inefficiencies that negatively impact financial performance. Private sector involvement improves project management efficiency, reduces operational costs, and improves financial returns over time (Ittmann, 2017). These efficiency gains improve infrastructure profitability and enhance Transnet's overall financial sustainability.

Long-Term Financial Sustainability and Strategic Financial Impact

Public-private partnerships contribute to long-term financial sustainability by improving infrastructure performance, increasing revenue generation, and reducing financial risk. Improved infrastructure capacity enables Transnet to increase cargo volumes, improve

operational efficiency, and enhance profitability. Havenga et al. (2023) emphasize that improving Transnet's infrastructure performance is essential to reducing logistics costs and improving South Africa's trade competitiveness. PPPs enable infrastructure expansion and operational improvements that support increased export capacity and economic growth.

By leveraging private capital, Transnet improves its financial sustainability without increasing debt levels or placing additional strain on public finances. PPPs therefore represent a strategic financial mechanism that improves infrastructure performance, enhances revenue generation, and strengthens Transnet's long-term financial position.

In sum, Public-private partnerships provide a financially sustainable solution for infrastructure development at Transnet Port Terminals. PPPs enable Transnet to access private capital, improve infrastructure efficiency, increase revenue generation, and reduce financial risk. Real-world examples, including the Richards Bay container facility concession and Durban Container Terminal partnership, demonstrate how private investment improves infrastructure capacity and financial performance (Engineering News, 2025; News24, 2026; SAnews, 2025). Academic research confirms that PPPs improve infrastructure delivery, operational efficiency, and financial sustainability by leveraging private sector investment and expertise (Havenga et al., 2023; Ittmann, 2017). By implementing PPPs as a financial strategy, Transnet Port Terminals can improve infrastructure performance, strengthen financial sustainability, and support long-term economic growth.

PDCA Cycle

To best ensure feasibility of our strategy and suggested Dual-Track implementation plan, the PDCA framework would be of assistance to ensure facts are primary in TPT's decision making (Koskela et al., 2019). We would plan to commence our strategy in a highly congested, weather affected port (such as Durban Pier 2) as this would best assist in the assessment of our strategy. This concentrated plan would also accommodate the reduction of risk and would be checked through the various KPI's outlined below. If proven to be successful TPT would be at liberty to act on a standardised roll out across various regions. Ultimately, resulting in a culture of continuous improvement within TPT as a consequence of continuous feedback loops across the entity (Koskela et al., 2019).

Key Performing Indicators

Financial KPIs

1. EBITDA Margins

EBITDA (Earnings Before Interest, Tax, Depreciation and Amortisation) margin measures core operational profitability before the effects of financing and accounting decisions. For TPT, this is particularly critical because Transnet's heavy debt burden and cross-subsidisation of loss-making divisions can distort bottom-line profitability. Monitoring EBITDA margins isolates TPT's true operational performance from Transnet's broader financial obligations, enabling regulators such as the Ports Regulator of South Africa and investors to assess

whether TPT's terminals are genuinely efficient or merely appearing profitable due to tariff inflation (Chasomeris & Fakir, 2020; Havenga et al., 2023).

2. Operating Profit Margins

Operating profit margin measures the percentage of revenue retained after all direct operating costs are deducted. For TPT, where chronic underinvestment has led to equipment failures, ageing infrastructure, and rising maintenance costs, this margin directly reflects the operational cost burden. A declining operating margin signals that inefficiencies such as vessel delays, crane breakdowns, and low berth productivity are eroding profitability. It therefore serves as a critical early warning indicator within any operational turnaround strategy (Havenga et al., 2023; Mthembu & Chasomeris, 2025).

3. Cargo Throughput in Tons

Cargo throughput, the total volume of cargo processed through a terminal over a given period, is arguably the most direct measure of a port's operational capacity and efficiency. For TPT, declining throughput at key terminals such as Durban Container Terminal directly translates to lost revenue, increased vessel waiting times, and the imposition of congestion surcharges by major shipping lines such as Maersk, CMA CGM, and MSC. Throughput data also underpins TPT's competitive positioning relative to regional ports such as Durban versus Ngqura, and internationally against benchmark ports (Notteboom et al., 2022; USDA/FAS, 2024). Monitoring throughput trends is therefore essential to evaluating whether operational interventions are yielding measurable results.

4. Return on Assets (ROA)

ROA measures how efficiently a company generates profit from its asset base, calculated as net income divided by total assets. In TPT's context, this is especially significant given the scale of capital tied up in port infrastructure much of which is ageing and underutilised. A low or declining ROA indicates that TPT's assets are not generating adequate returns, which in turn weakens the investment case for infrastructure modernisation and deters private capital participation. Given that Transnet's total capital investment declined from R33 billion in 2015 to R13 billion in 2022, ROA serves as a key metric to demonstrate to potential investors and the World Bank whether asset recapitalisation programmes are translating into improved financial returns (Havenga et al., 2023; World Bank, 2025).

Non-financial KPI's

1. Ship turnaround time (STAT)

Ship turnaround time is a key efficiency indicator with major international carriers considering it as central in selecting their port of choice. It involves all activities from the moment of anchorage until the ship departs, therefore this KPI tends to reflect broader issues of backlogs, congestion, and idle time wasted. It indicates the extent to which a port can claim to be efficient and TPT's current STAT of 68 hours is far behind the global average of

24 hours. Increased efficiency and less waste will be reflected in a decrease from 68 hours to 50 hours in the long-term (Dlamuka, 2022).

2. Average Dwell Time

Separated into cargo dwell time and port dwell time. The former refers to the time between vessel arrival and container exit from the ports, and the latter denotes the time spent by ships waiting within the port. High average dwell times reflect slow overall movement of containers and emphasise the need for faster, more efficient, cargo handling. When measured against other Sub-Saharan African (SSA) countries, South Africa has an impressive dwell time of between 3-5 days, averaging 3.93 days according to PRSA (2016). However, there is a need to reduce it if TPT aims to compete globally.

3. Ship Anchorage Waiting Time

An indicator of port congestion as it measures the hours a vessel waits outside a port before it can come into a berth for cargo work to begin. Dlamuka (2022) indicates that the main source of long waiting times is a lack of immediate berths available thus emphasising the need for infrastructure in decreasing wait times. Addressing this aspect will aid in gaining a competitive advantage over other SSA ports.

4. Berth Occupancy Rate (BOR)

Refers to the total time vessels occupy berthing facilities and is calculated by dividing the total time a vessel spends at the berth by the total berth hours available. This is one of the KPIs that requires more rigorous monitoring as TNPA numbers broadly estimate the berth occupancy as between 55% to 85% (Dlamuka, 2022). The lower end of the estimate would indicate excess and un-utilised capacity, whereas over-utilisation (65+) indicates congestion. New data monitoring systems would be able to track this more carefully to better understand the core of berth inefficiency and address it more directly.

Conclusion

This report set out to provide an informed operations strategy that ushers TPT into alignment with its business objectives. Systemic challenges, such as logistical bottlenecks and mismanagement, have resulted in profound implications for South Africa's structural competitiveness in the global port freight landscape. Moreover, these challenges have negatively impacted South Africa's national GDP, resulting in an urgent need to shift TPT from a reactive strategy of operation to a proactive, technologically-centric operations strategy. We have proposed the implementation of Campbell Scientific Automated Weather Systems (AWS), SAP S/4HANA and Navis N4 in our strategy as integral technological systems. These aid in addressing identified challenges in TPT pertaining to their outdated legacy systems, wait-waste and operational inefficiencies. The use of these systems would ensure greater efficiency, reduction of vessel waiting times and an improvement in terminal throughput. Additionally, the use of a Dual-Track Approach for implementing the strategy would ensure greater stabilisation without compromised transformation within the long-term,

at TPT. Moreover, this strategy addressed the Triple Bottom Line as a reduction in idling time would contribute to greater environmental sustainability through the reduction of maritime carbon emission. Simultaneously, the proposal for workforce upskilling would ensure social sustainability and would allow for the transformation of TPT into a more agile, data-driven organisation. Ultimately, we believe our proposed strategy and its successful execution would result in improving South Africa's economic reality, a reduction in associated costs operating business in South Africa and create a better customer experience for those reliant on Transnet Port Terminals.

Reference List:

- Annanki, S. (2025a). Digital transformation nexus: SAP S/4HANA's impact on organizational change and societal progress. *Journal of Computer Science and Technology Studies*, 7(10), 85–94. <https://doi.org/10.32996/jcsts.2025.7.10.9>
- Annanki, S. (2025b). SAP S/4HANA Cloud: Driving agility, innovation, and growth in ERP. *European Journal of Science, Innovation and Technology*, 5(4), 192–203. <https://ejst-journal.com/index.php/ejsit/article/view/710>
- Arena, L., Mola, L., Remond, N., & Rowe, F. (2020). How do enterprise software providers adapt their strategies to the cloud? An analysis through SAP Hana journey based on the evolution of SAP's discourse (2010–2018). In *Proceedings of the 53rd Hawaii International Conference on System Sciences* (pp. 1–10).
- Austin, G. (2011). *Assessment of freight train derailment risk reduction measures: Part A final report*. Det Norske Veritas (DNV) Ltd.
- Ayyalasomayajula, M. M. T. (2022). Multi layer SOMs for robust handling of tree structured data. *International Journal of Intelligent Systems and Applications in Engineering*, 10(2), 275. Retrieved from: <https://ijisae.org/index.php/IJISAE/article/view/6937>
- BDO Advisory South Africa. (2023). *The economic toll of port and rail inefficiencies*. <https://www.bdo.co.za>
- Campbell Scientific. (n.d.). *Automatic Weather Stations (AWS) and meteorologic instruments: Worldwide standard for climate and boundary-layer meteorology*. <https://www.campbellsci.co.za/aws-meteorology>
- Chiranga, M. R. (2018). An investigation into the causes of failures in railway. *International Journal of Sustainable Development & World*, 9(1), 1–10.
- Christopher, M. (2016). *Logistics and supply chain management* (5th ed.). Pearson Education.
- Cope, L. (2025, November 9). *What is railway rolling stock? Types and engineering*. Engineer Fix. <https://engineerfix.com/what-is-railway-rolling-stock-types-and-engineering/>
- Daily Investor. (2023). *Transnet crisis will cost South Africa 5% of GDP*. <https://www.dailyinvestor.com>
- Daily Maverick. (2023). *Transnet's financial crunch intensifies*. <https://www.dailymaverick.co.za>
- Dale, B. G., Bamford, D., & van der Wiele, T. (2016). *Managing quality: An essential guide and resource gateway* (6th ed.). Wiley.

de Langen, P. W., Meijeren, J. V., & Lakkien, M. (2023). The strategy of the port development company: A business ecosystems perspective — Port of Rotterdam. *Research in Transportation Business & Management*, 47, Article 100911. <https://doi.org/10.1016/j.rtbm.2022.100911>

Dean, P. (2024, October). Transnet Freight Rail plans for future growth: South African national operator Transnet Freight Rail has faced several challenging years, but now appears to be making a recovery. *International Railway Journal*, 64(10), 28–31. <https://link.gale.com/apps/doc/A816617090/AONE?u=anon~58d0300&sid=googleScholar&xid=a0281b16>

Dlamuka, M. T. (2022). Key performance indicators for container ports: A case of Weighted Efficiency Gains from Operations (WEGO) in South Africa (pp. 22-27) <https://researchspace.ukzn.ac.za/handle/10413/21260>

Flynn, B. B., Huo, B., & Zhao, X. (2010). The impact of supply chain integration on performance. *Journal of Operations Management*, 28(1), 58–71. <https://doi.org/10.1016/j.jom.2010.01.001>

George, T. M. (2018). *A review on the current condition of rail* [Master's thesis, University of Pretoria]. University of Pretoria Repository. <https://repository.up.ac.za/server/api/core/bitstreams/8fd6a95b-2438-49f6-b7bf-abbbe1bab1f6/content>

Goswami, M. J. (2020). Leveraging AI for cost efficiency and optimized cloud resource management. *International Journal of New Media Studies: International Peer Reviewed Scholarly Indexed Journal*, 7(1), 21–27.

Grabski, S. V., Leech, S. A., & Schmidt, P. J. (2011). A review of ERP research. *International Journal of Accounting Information Systems*, 12(1), 1–26. <https://doi.org/10.1016/j.accinf.2010.03.007>

Havenga, J. H., Simpson, Z. P., King, D., & de Bod, A. (2023). The macrologic effect of a state-owned enterprise, Transnet, on the South African economy. *Journal of Transport and Supply Chain Management*, 17, 1–12. <https://doi.org/10.4102/jtscm.v17i0.888>

Hellenic Shipping News. (2023). *South Africa's ports and rail system facing imminent 'collapse'*. <https://www.hellenicshippingnews.com>

Koskela, L., Tezel, A., & Patel, V. (2019). Theory of quality management: Its origins and history. In C. Pasquire & F. R. Hamzeh (Eds.), *Proceedings of the 27th Annual Conference of the International Group for Lean Construction (IGLC)* (pp. 1381-1390). <https://doi.org/10.24928/2019/0259>

Lugt, L. V. D., & de Langen, P. W. (2023). Advancing public interests through state ownership: The case of Port of Rotterdam. *GeoJournal*, 88, 4133–4151. <https://doi.org/10.1007/s10708-023-10842-8>

Maritime and Port Authority of Singapore. (n.d.). *Port infrastructure and terminals overview*. <https://www.mpa.gov.sg>

Moyo, A. (2025, September). *Transnet upgrades ERP system to SAP S/4HANA*. SAP Africa News Center. <https://news.sap.com/africa/2025/09/transnet-upgrades-erp-system-to-sap-s-4-hana/>

Nidumolu, L., & Prahalad, C. (2024). Digital transformation and enterprise sustainable development. *Finance Research Letters*, 60, Article 104902. <https://doi.org/10.1016/j.fl.2023.104902>

Nigel Slack, S. C. (2010). *Operations Management (6th ed.)*. Harlow, England: Pearson Education Limited.

Norwegian Safety Investigation Authority. (2021). *Framework and analysis process for systematic safety investigations*. Norwegian Government.

Notteboom, T., Pallis, A., & Rodrigue, J.-P. (2022). *Port economics, management and policy*. Routledge. <https://porteconomicsmanagement.org>

Oracle. (n.d.). *What is ERP?* <https://www.oracle.com/applications/erp/what-is-erp.html>

Pallis, A., & Rodrigue, J.-P. (2020, November 24). *Chapter 10.2: Port efficiency*. Port Economics, Management and Policy. <https://porteconomicsmanagement.org/pemp/contents/part10/port-efficiency/>

Pelphrey, M. W. (2015). *Directing the ERP implementation: A best practice guide to avoiding program failure traps while tuning system performance*. CRC Press.

PGIM Private Capital. (2023). *Port of Rotterdam partner story — €125m private placement*. <https://www.prudentialprivatecapital.com>

PR Newswire. (2023). *South Africa ports and harbours maintenance and operation industry report 2023*. <https://www.prnewswire.com>

PSA International. (2024). *Corporate overview*. <https://www.globalpsa.com>

Public-Private Infrastructure Advisory Facility. (n.d.). *Module 3: Alternative port management structures and ownership models — Port functions, services, and administration models*. World Bank. <https://www.ppiaf.org>

Railway Safety Regulator. (n.d.). *State of safety 2017/2018*. <https://www.rsr.org.za/wp-content/uploads/2024/02/RSR%20State%20of%20Safety%20Report%202017%20-%202018>

Sankar, C. S., & Rau, K. (2006). *Implementation strategies for SAP R/3 in a multinational organization: Lessons from a real-world case study*. Cybertech Publishing.

South African Association of Freight Forwarders. (2023, November). *Port congestion reports*. IOL Business. <https://www.iol.co.za>

Tamela Corporate Advisory. (2024). *The real cost of infrastructure bottlenecks in the South African economy*. <https://www.tamela.co.za>

The Business Year. (2024). *Port series: Durban, South Africa*. <https://www.thebusinessyear.com>

Transnet. (n.d.). *Transnet Foundation*. <https://www.transnet.net/TransnetFoundation>

Transnet. (2017). *Our history*. <https://www.transnet.net/OurHistory>

Transnet. (2023a). *Overview*. <https://www.transnet.net/Overview>

Transnet. (2023b). *Reinvent for growth: Transnet recovery plan*. <https://www.transnet.net/getFile.ashx?id=5741>

Transnet. (2024). *Transnet port terminals*. <https://www.transnet.net/TransnetPortTerminals>

Transnet National Ports Authority. (2019). *Emergency preparedness & response*. <https://www.transnetnationalportsauthority.net/DoingBusinesswithUs/SafetyHealthEnvironment/Pages/Emergency-Preparedness-and-Response.aspx>

Transnet Port Terminals. (2019). *ICT at Transnet port terminals*. <https://www.transnetportterminals.net/About/Pages/ICT.aspx>

Transnet Port Terminals. (2023). *Annual report 2022/23*. <https://www.transnet.net>

Transnet Port Terminals. (2024). *Annual report 2023/24*. <https://www.transnet.net>

Transnet SOC Ltd. (2024). *Integrated annual reports 2019–2024*. <https://www.transnet.net>

Tsang, A. H. C. (2002). Strategic dimensions of maintenance management. *Journal of Quality in Maintenance Engineering*, 8(1), 7–39. <https://doi.org/10.1108/13552510210420577>

UN Centre for Trade and Development. (n.d.). *United Nations resilient maritime logistics: About the guidebook*. <https://resilientmaritimelogistics.unctad.org/guidebook/about-guidebook>

United Nations Economic and Social Commission for Western Asia. (2022). *Public-private partnership for ports development and operation*. <https://www.unescwa.org>

USDA Foreign Agricultural Service. (2024). *Port challenges cause ships and export opportunities to pass South Africa by*(GAIN Report No. SF2024-0005). <https://www.fas.usda.gov>

Wang, W., & Li, G. (2011). Category of derailment mechanism and prevention for high-speed vehicle. *Journal of Transportation Engineering*, 137(10), 730–737. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000262](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000262)

Wikipedia. (2024). *Port of Singapore Authority (PSA) — History and development*. NLB Singapore Infopedia. <https://eresources.nlb.gov.sg>

Wipro. (n.d.). *Enhancing enterprise sustainability with SAP*. <https://www.wipro.com/applications/enhancing-enterprise-sustainability-with-sap/>

World Bank. (2022). *Container port performance index (CPPI) 2022*. <https://www.worldbank.org>

World Bank. (2023a). *Container port performance index (CPPI) 2023*. <https://www.worldbank.org>

Heilig, L., Lalla-Ruiz, E., & Voß, S. (2017). Digital transformation in maritime ports: Analysis and a game theoretic framework. *Netnomics*, 18(2–3), 227–254. <https://doi.org/10.1007/s11066-017-9122-x>

Notteboom, T., & Rodrigue, J.-P. (2005). Port regionalization: Towards a new phase in port development. *Maritime Policy & Management*, 32(3), 297–313. <https://doi.org/10.1080/03088830500139885>

Slack, N., & Brandon-Jones, A. (2019). *Operations management* (9th ed.). Pearson Education.

World Bank. (2023b). *South Africa: Infrastructure and logistics assessment*. <https://www.worldbank.org>

Appendices

Appendix A: Table Showing the Geographic Landscape of TPT

Port	Position	Cargo	Strategic importance	Operations Systems
Richards Bay	Northern KZN	Mineral bulk Break bulk	Bulk port of choice in Southern Africa	Computer controlled network of conveyer belts to transport cargo between the quayside and manufacturers (Dry bulk only) Break bulk as a skip-loading that relies on Road Motor Transport (RMT)
Durban	KZN, 50 kms away from King Shaka Airport	Break bulk Agri bulk Containers RORO Automotive	Handles 60% of SA containers Serves a large part of South and Southern Africa	Truck Appointment System All cargo remains on wheels throughout Centralised management cluster, terminal in and out processing facilities
East London	Eastern Cape	RORO Break bulk Agri bulk Containers		River port RORO system with a multi-level car terminal
Ngqura	Eastern Cape	Containers	Special focus on manganese handling	Automated Gate System (AGS) with pre-booking functionality
Cape Town	Western Cape	MPT	Second largest seaport in SA Ideal location for trade with South America and the Far East	
Saldanha	Western Cape	Mineral bulk Break-bulk		Port is separated into an MPT and an iron ore specific terminal
Gqeberha	Eastern Cape	Mineral bulk Break bulk Containers RORO	Currently exploring a niche market of tugboat and navy building (potential comparative advantage)	Little specialised tech systems

Sourced from: (US International Trade Dev & Transnet, 2023)

Appendix B: TPT Division Financial Performance

Financial Year	TPT Revenue (R billion)	TPT EBITDA (R billion)	EBITDA Margin (%)	ROIC (%)	Key Volume Indicators
2021/22	~14.5	4.2	29%	15%	Container volumes 4.1m TEU
2022/23	16.8	5.9	35%	>22%	Automotive +20.9%; Containers –2.4%; Bulk –1.5%
2023/24	18.5	6.33	34%	22%	Automotive 826,033 units; Revenue +9.7%
2024/25	10.5	-3.5	-33%	<20%	Volume Decrease in Containers & Bulk

Sourced from: (Transnet, n.d)

Appendix C: Macro-Economic Cost on GDP and Trade

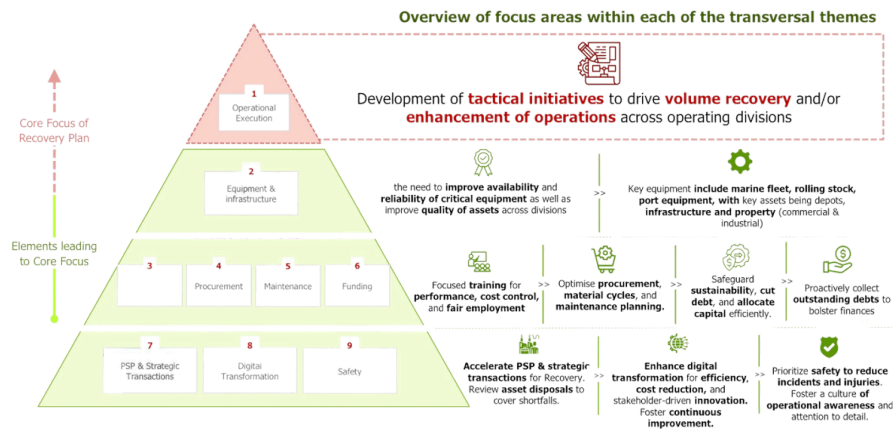
Metric	2019	2022	Change / Impact
Total port volumes (tons)	R215.1 million	R173.1 million	–19.5% decline
Infrastructure cost to economy (direct)	R9.7bn (0.17% GDP)	R15.5bn (0.23% GDP)	+60% increase
Infrastructure cost (incl. opportunity cost)	R84.5bn (1.5% GDP)	R458.4bn (6.7% GDP)	+442% increase
Logistics Performance Index ranking	Top 20 globally	19th globally	As of 2024 , TPT ports rank amongst the lowest globally
Port efficiency change (2010–2019)	Baseline	–15% (vs +30% for facilitated SOEs)	Structural competitiveness loss

Sourced from: Journal of Transport and Supply Chain Management (2023); Tamela Advisory (2024); Daily Investor (2023); Investec Head of Logistics, Newzroom Afrika (2023).

Appendix D: Financial loss to TPT

Incident / Period	Estimated Direct Loss	Source
Oct–Dec 2023: Durban congestion crisis	R160m+ lost revenue (TPT estimate)	Daily Maverick, Nov 2023
Oct 2023: Pier 2 weather/equipment failures	159 hours lost operational time; 20+ vessels waiting	TPT Operations Data
Nov 2023: Peak crisis	63 vessels at anchorage; 61,000+ containers stuck at sea	SAAFF / IOL, Nov 2023
Daily cost to economy (direct + indirect)	R124m–R263m per day	SAAFF, Nov 2023
Total port & rail cost (18 months, 2022–23)	R150bn estimated	BDO Advisory, 2023
Oct 2022 strike	R6bn economic loss	Engineering News, 2022

Appendix E: Recovery Plan (2023)



Sourced from: Transnet, 2023b